Ex 8: Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Topic: Patient Appointment Booking System

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Note: this report sets out the RBAC configuration for the application as a design. The settings should be checked against the live application before being relied on.

1. Aim

To configure user management and role-based access control in the Patient Appointment Booking application, so that each kind of user — administrator, doctor, receptionist and patient — can see and change only the records and components their work needs, using Zoho Creator's permission sets, roles and data sharing rules.

2. Procedure

Step 1: Open the security settings — In the application builder, switch to Edit mode and open Settings. Under the Permissions group, choose User Permissions, which has the Permissions, Roles and Data Sharing tabs.

Step 2: Define permission sets — On the Permissions tab, create a permission set for each type of user (Receptionist, Doctor and Patient) and choose which forms, reports and pages each one may open and what it may do with them.

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Step 3: Build the role hierarchy — On the Roles tab, arrange the roles as a tree with Hospital Admin at the top and Doctor, Receptionist and Patient below it, so that access to records flows upward through the hierarchy.

Step 4: Add users — From the Users area, invite each person by email and assign one permission set and one role to each user.

Step 5: Create a data sharing rule — On the Data Sharing tab, add a rule that shares the booking records captured by receptionists with doctors, choosing the form, the sharing roles and the level of access.

Step 6: Save and verify — Save the rule and reopen it to confirm that every setting was stored as intended, then sign in as each type of user to check what they can see.

3. Output

3.1 Permission Sets

Permission Set	Intended For	Access to the Booking Form and Reports
Admin	Hospital administrator	Full access to all forms, reports, pages and settings
Receptionist	Front-desk staff	Create, view and edit bookings; view the daily schedule report
Doctor	Consulting doctors	View bookings assigned to them; no delete or export
Patient	Patients using a portal	Submit a booking and view their own bookings only

Table 1: Proposed Permission Sets

3.2 Role Hierarchy

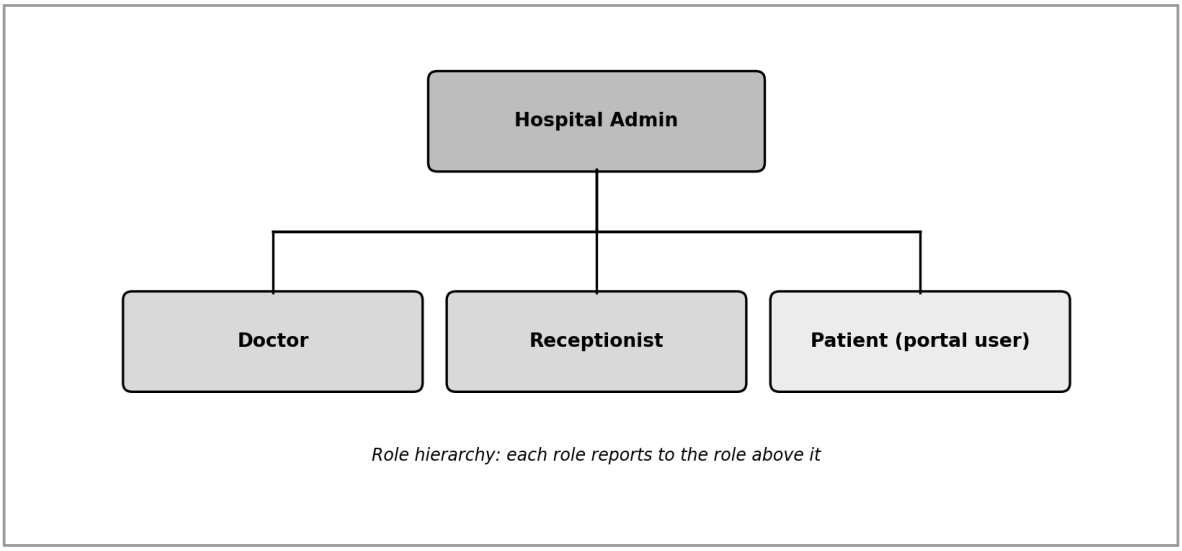


Figure 1: Proposed Role Hierarchy (illustrative diagram)

Hospital Admin sits at the top and can reach the records of every role beneath it. Doctor, Receptionist and the portal Patient role are siblings under it, so none of them can see another's records unless a sharing rule allows it.

3.3 Data Sharing Rule

Setting	Value
Rule name	Receptionist_to_Doctor
Form	Patient_Appointment_booking
Records shared from	Role: Receptionist
Records shared to	Role: Doctor
Level of access	Read only

Table 2: Proposed Data Sharing Rule



Figure 2: Schematic of the Proposed Data Sharing Rule (illustrative diagram, not a screenshot)

With this rule, bookings entered by the receptionist become visible to doctors without giving doctors permission to edit them. If doctors also need to update the status of a booking, the level of access can be raised to Read/Write.

3.4 Resulting Access Matrix

Area	Admin	Receptionist	Doctor	Patient
Booking records	Full control	Create, view, edit	View (shared)	Own bookings only
Contact details	View, edit	View, edit	View	Own details
Reports and dashboard	All reports	Daily schedule	Own schedule	No access
Workflows and scripts	Manage	No access	No access	No access
Users and roles	Manage	No access	No access	No access

Table 3: Access by User Type

3.5 How the Setup Is Verified

Check	Expected Result
Sign in as Receptionist and open the booking form	Can add and edit bookings; cannot open settings
Sign in as Doctor and open the booking report	Can see bookings shared by the Receptionist; cannot edit or delete them
Sign in as Patient through the portal	Can submit a booking and sees only their own entries
Open the saved sharing rule again	All fields show the values configured in Table 2

Table 4: Verification Checks

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4. Result

A role-based access control design was prepared for the Patient Appointment Booking application, covering permission sets for each type of user, a role hierarchy under the hospital administrator, and a data sharing rule that lets doctors see bookings entered by receptionists. These settings are a proposed configuration and should be applied and tested in the live application, using the checks in Table 4.

Appendix: Zoho Creator Screenshots

Reference screenshots supplied from the teammate report for the same project. Use these as visual references and confirm that the settings match your own live application before submission.

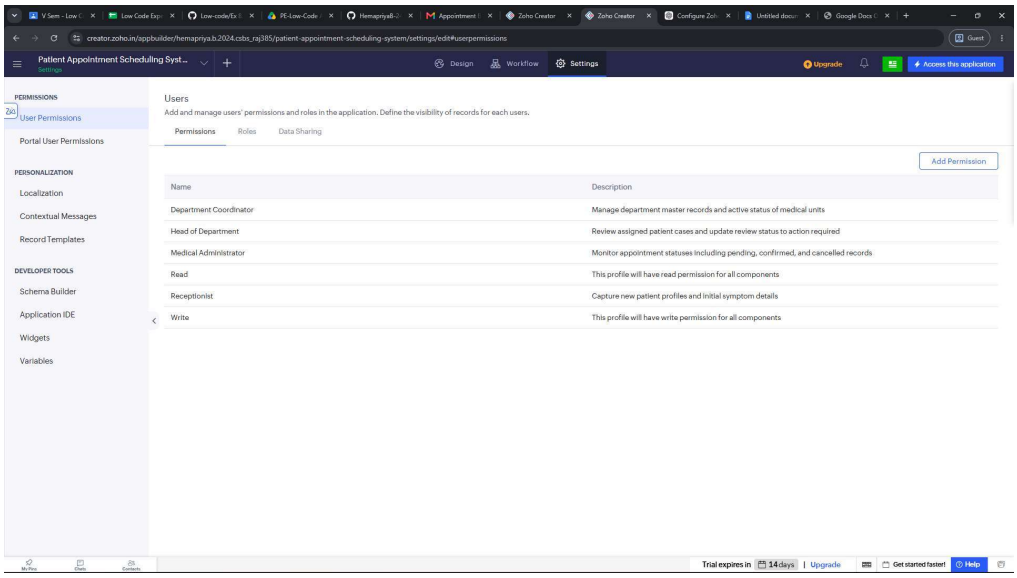


Figure A1. User Permissions — configured role/permission list in Zoho Creator.

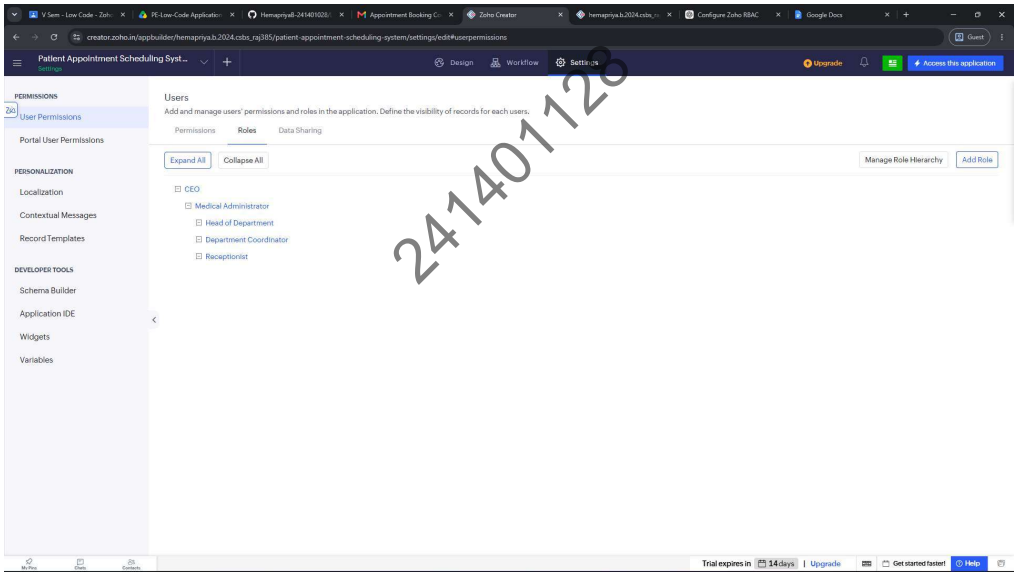


Figure A2. Role hierarchy — example hierarchy configured in the application.

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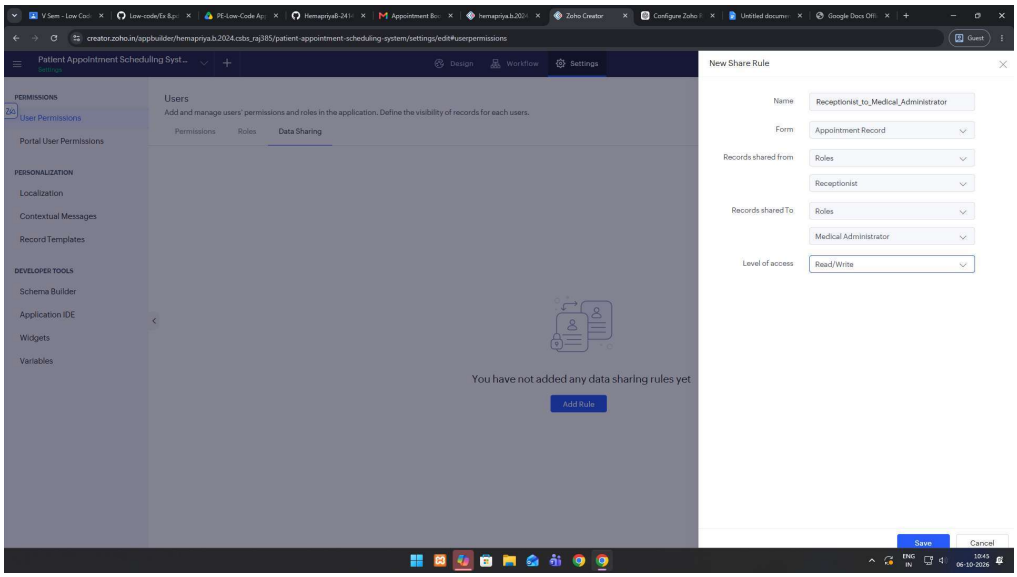


Figure A3. Data-sharing rule — new rule configuration screen.

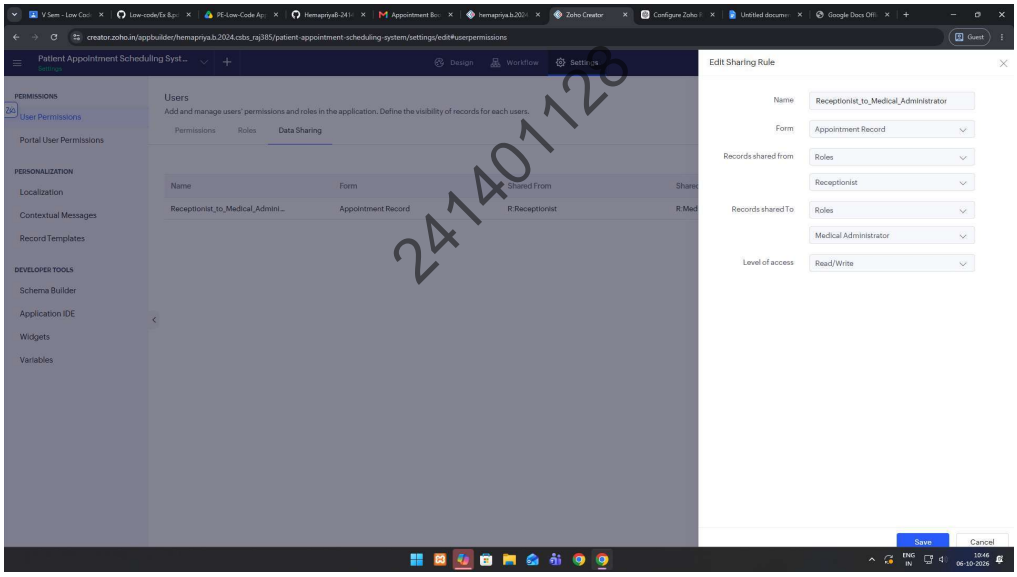


Figure A4. Data-sharing rule — saved rule details.

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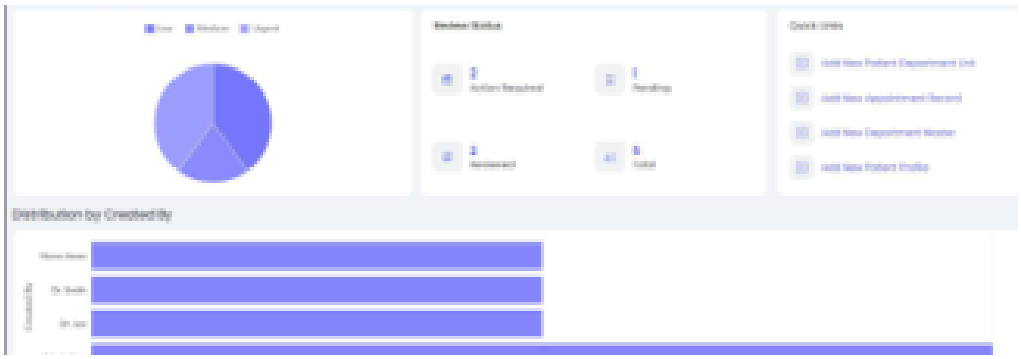


Figure A5. Application dashboard — example Patient Appointment Scheduling System interface.

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